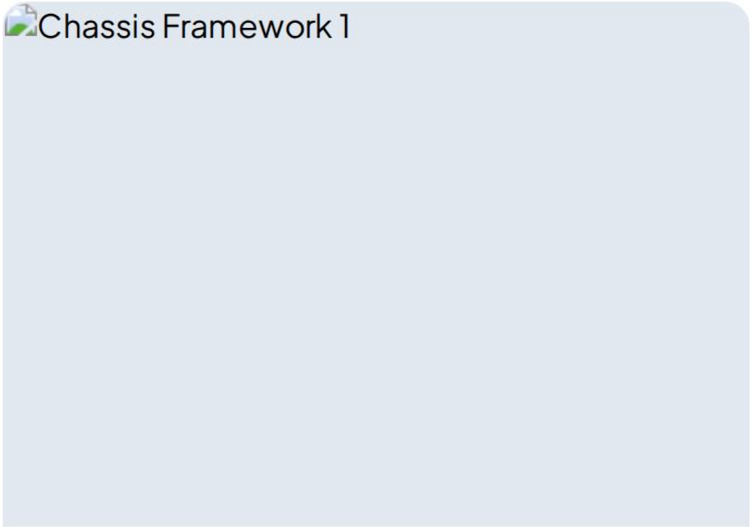
ROBOTICS ENGINEERING STUDIES

Components of the Mechanical Block of a Robot

Detailed breakdown of physical structures, chassis, joints, actuators, end effectors, and controllers.

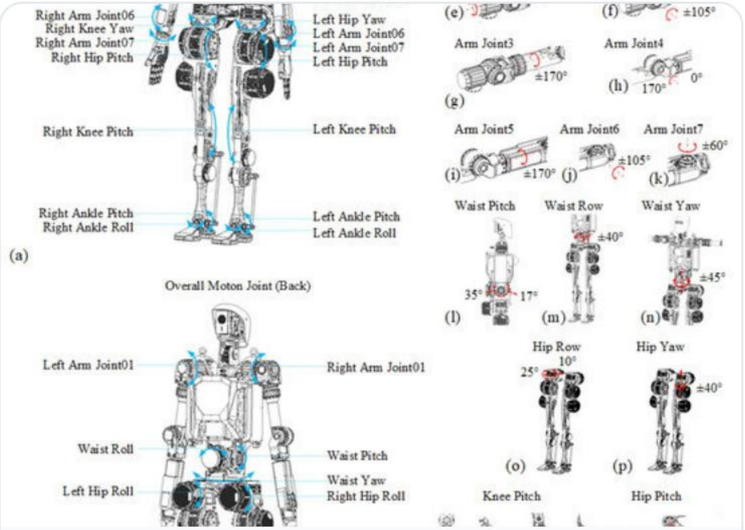
COMPONENT 01

Outer Body & Chassis Framework



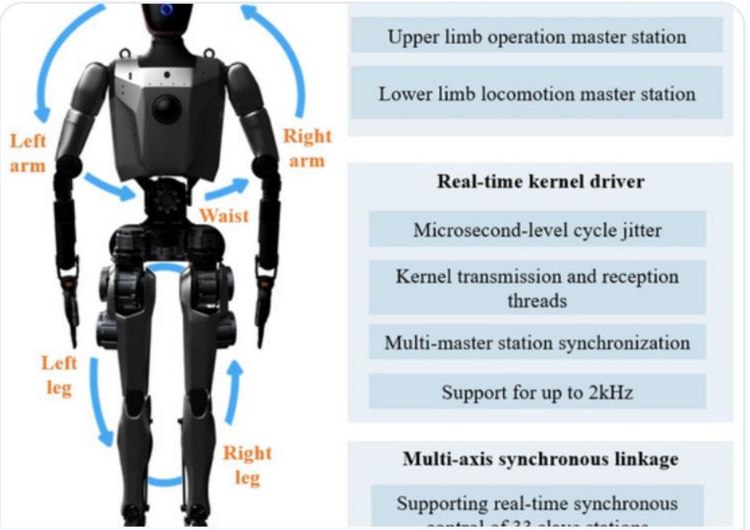
Physical Framework

Creates the tangible chassis supporting all components.



Form Integrations

Varies from wheeled robots to humanoids and cleaning units.

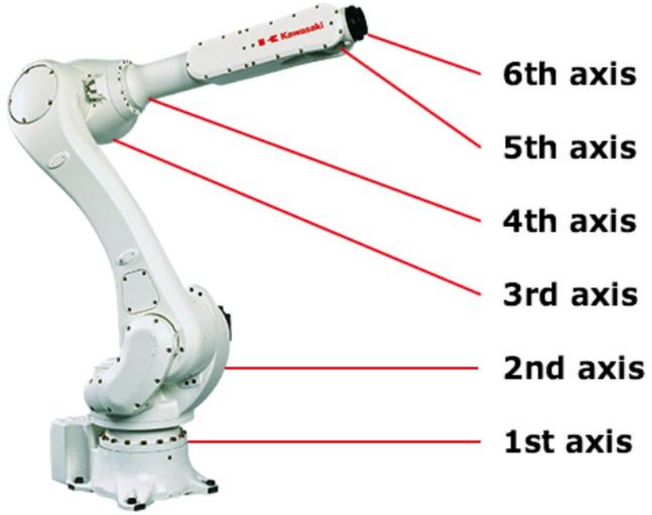


Application Look

Provides the unique appearance and protective outer skin.

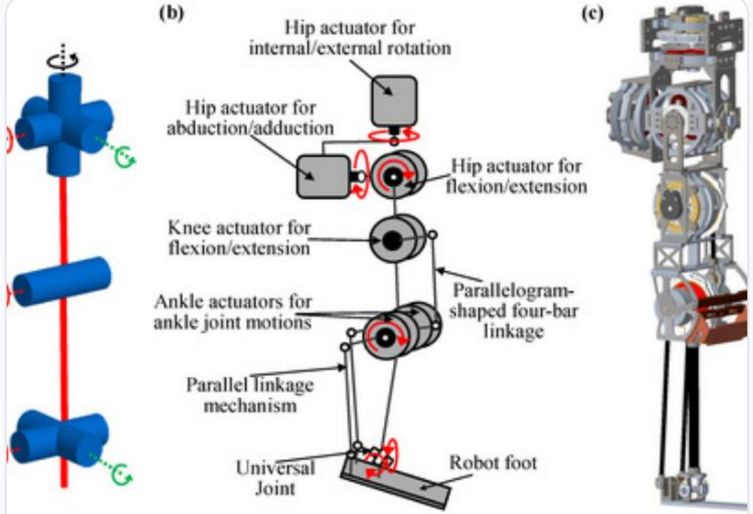
COMPONENT 02

Joints & Links Structure



Robot Mobility

Joints help robots move and articulate cleanly.



Structural Connectors

Links connect different parts of the robotic skeleton.



Combined Foundation

Joints and links work together to form the core structure.

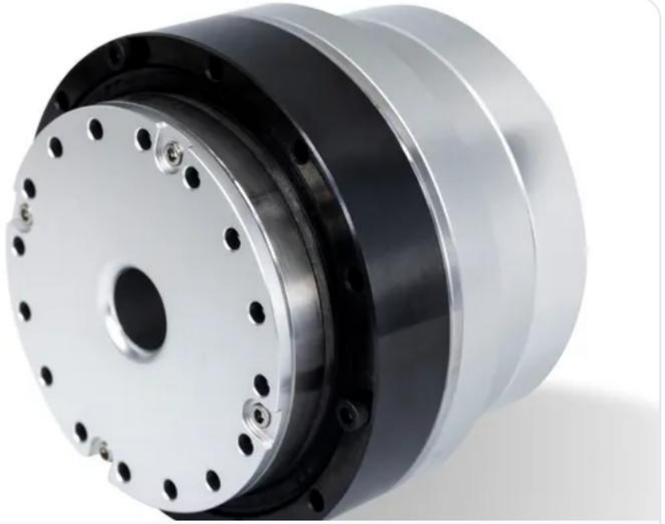
COMPONENT 03

Actuators & Drive Motors



Electric Motors

Standard rotary and linear electric motor drives.



Hydraulic Devices

Fluid-powered mechanisms used to drive heavy joints.



Harmonic Drive
Robot Joint Actuator

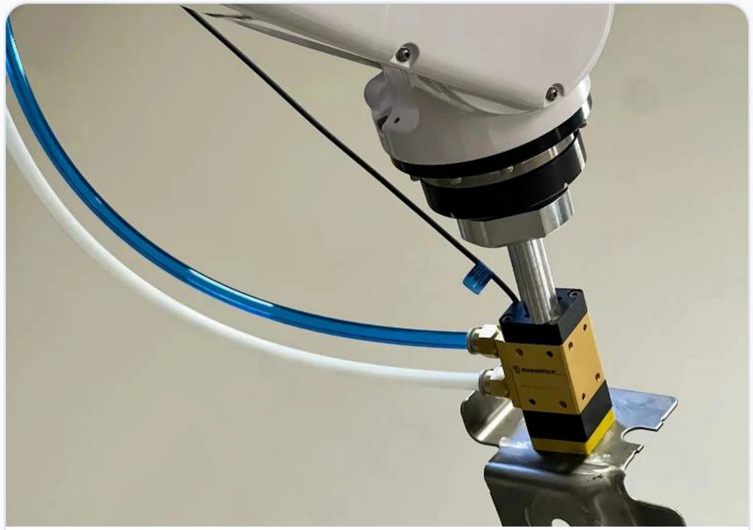
Planetary Reducer
Robot Joint Actuator

Motion Control

Enables precise movement across multiple axes.

COMPONENT 04

End Effectors & Tooling



Environment Interaction

Tools used to interact directly with objects and surroundings.



Gripping & Cutting

Grippers, cutters, and industrial welding torches.



Painting & Mopping

Specialized application heads for finishing and cleaning.

COMPONENT 05

Controller Units & Signaling

Controller 1

System Control

Controls the entire mechanical block safely.



Sensor Inputs

Processes inputs received from external sensors.



Signal Routing

Sends coordination signals to actuators and other blocks.

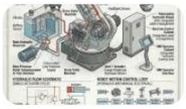
SUMMARY

The Anatomy of Robotic Mechanics

Chassis frameworks, joint articulation, precise actuators, and controllers combine to form robust robotic systems.

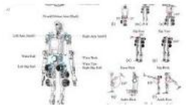
Outer Body | Joints & Links | Actuators | End Effectors | Controllers

Image Sources



https://lookaside.fbsbx.com/lookaside/crawler/media/?media_id=2191943241647386

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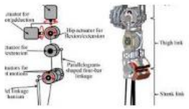
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Image Sources



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